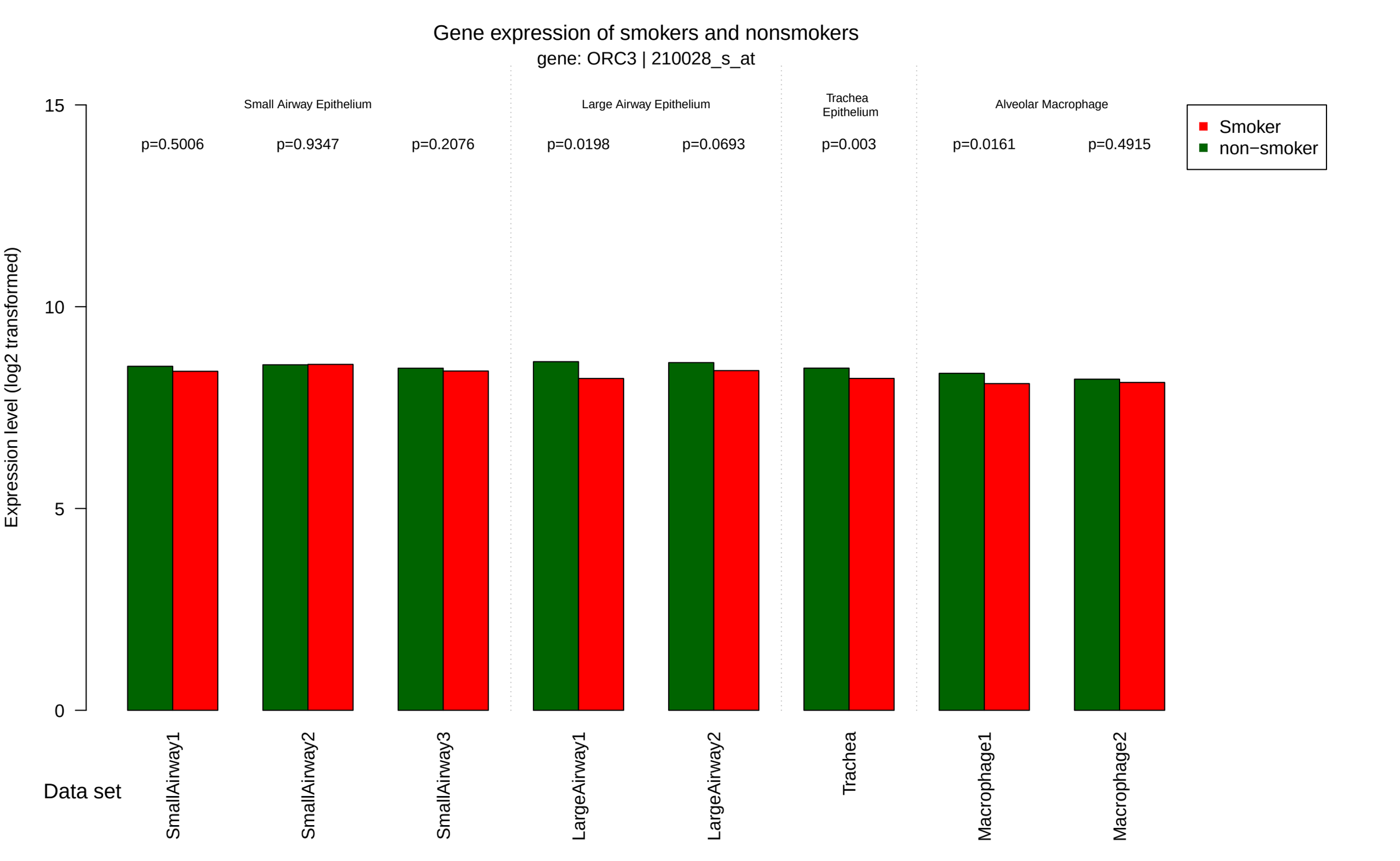
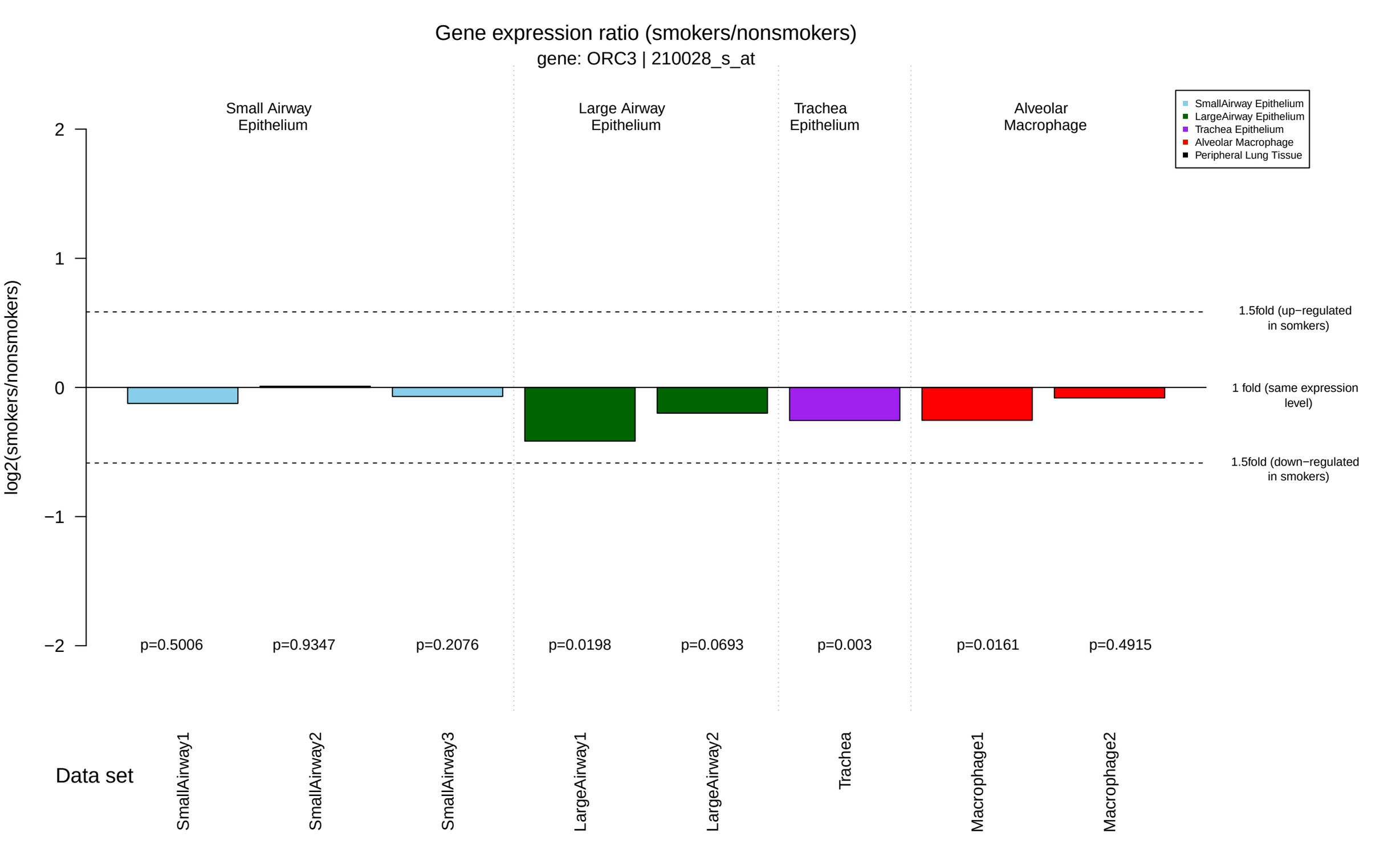
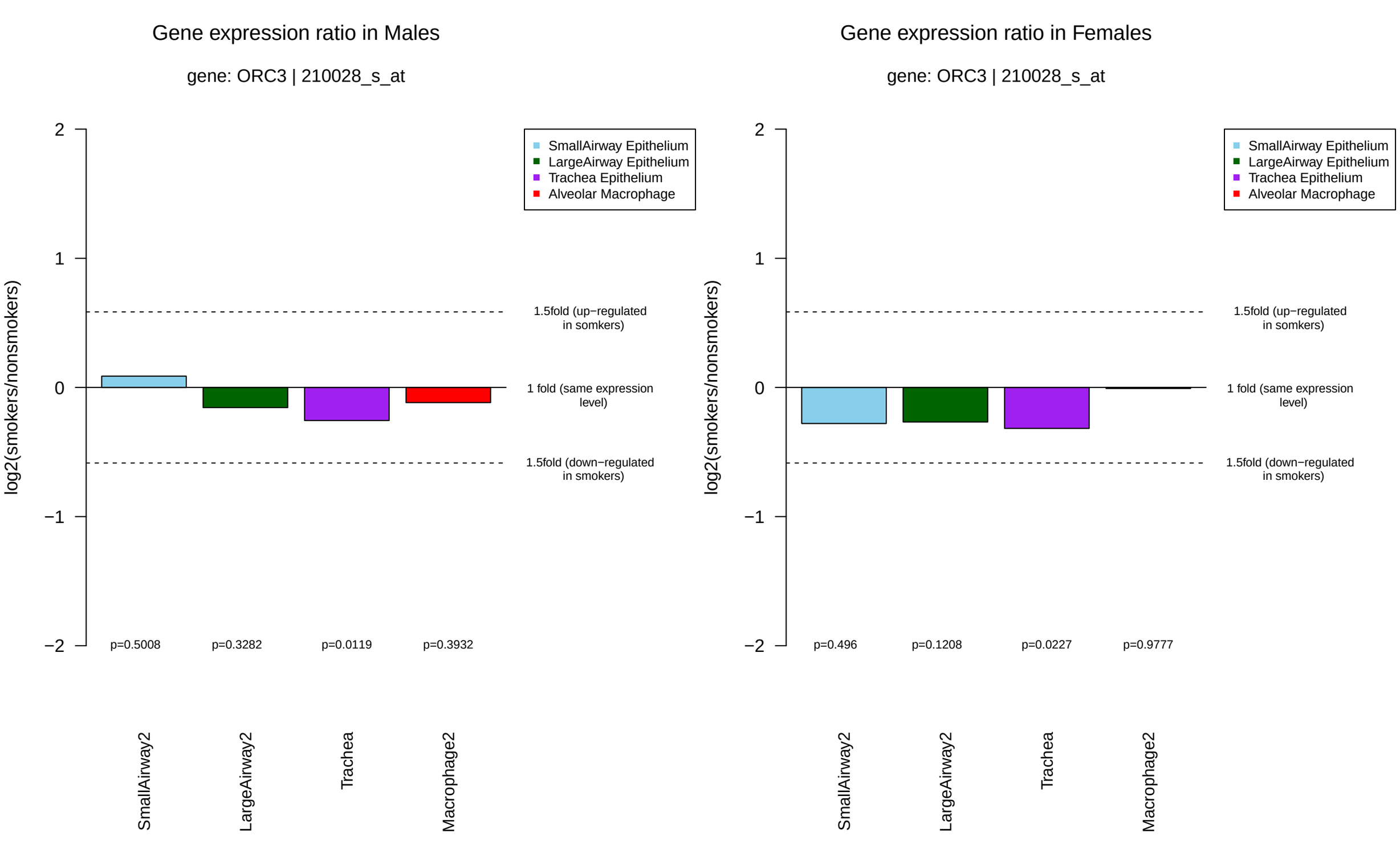
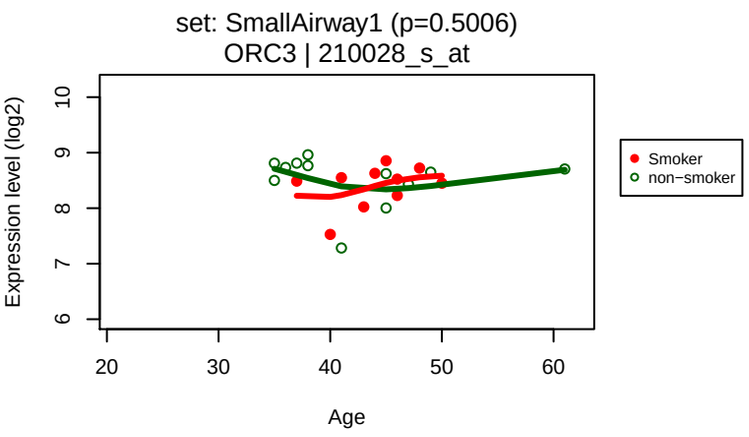
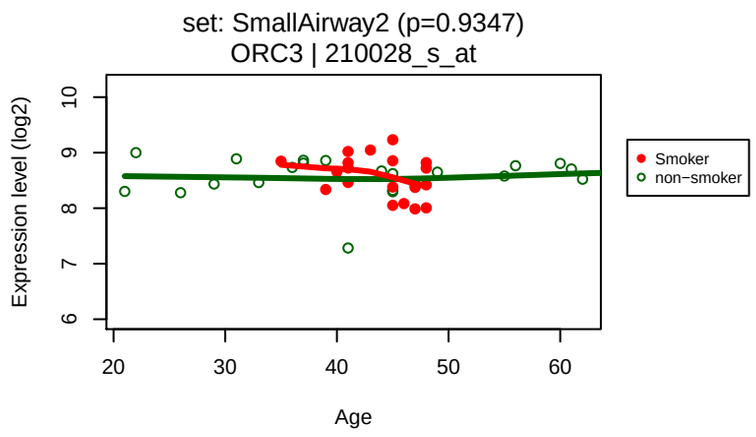
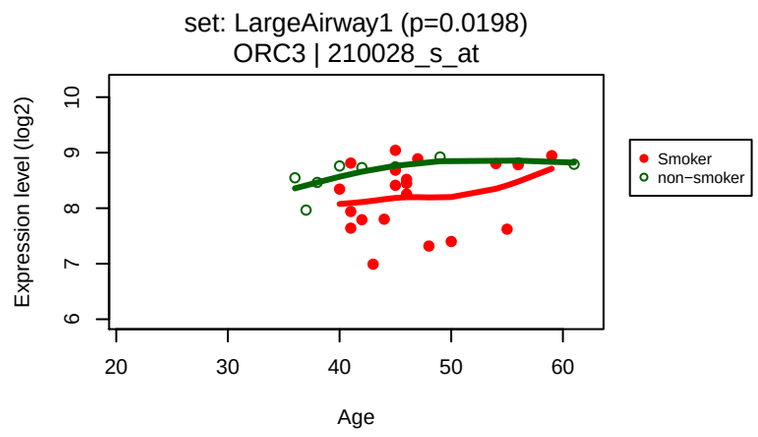

Gene expression of smokers and nonsmokers
gene: ORC3 | 210028_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5006
p=0.9347
p=0.2076
p=0.0198
p=0.0693
p=0.003
p=0.0161
p=0.4915
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

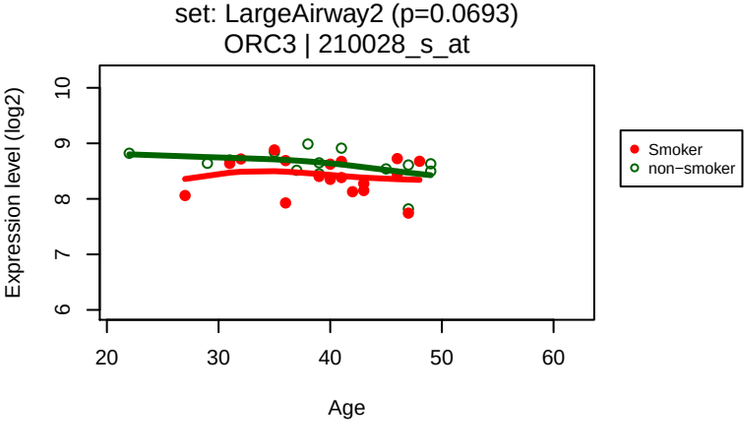

Gene expression ratio (smokers/nonsmokers)
gene: ORC3 | 210028_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5006
p=0.9347
p=0.2076
p=0.0198
p=0.0693
p=0.003
p=0.0161
p=0.4915
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

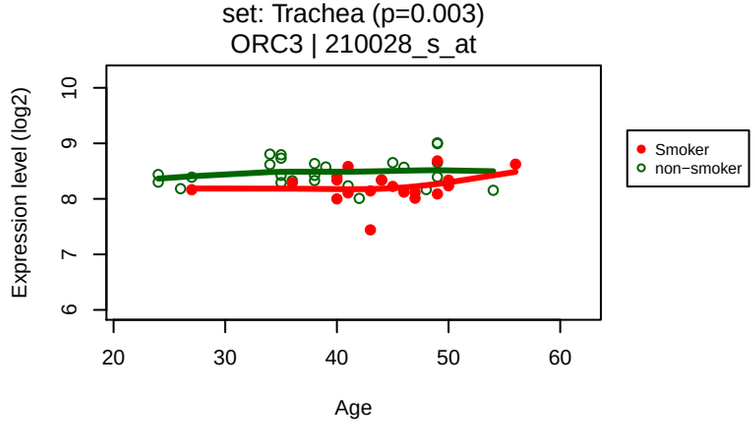

Gene expression ratio in Males
gene: ORC3 | 210028_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5008
p=0.3282
p=0.0119
p=0.3932
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ORC3 | 210028_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.496
p=0.1208
p=0.0227
p=0.9777
SmallAirway2
LargeAirway2
Trachea
Macrophage2

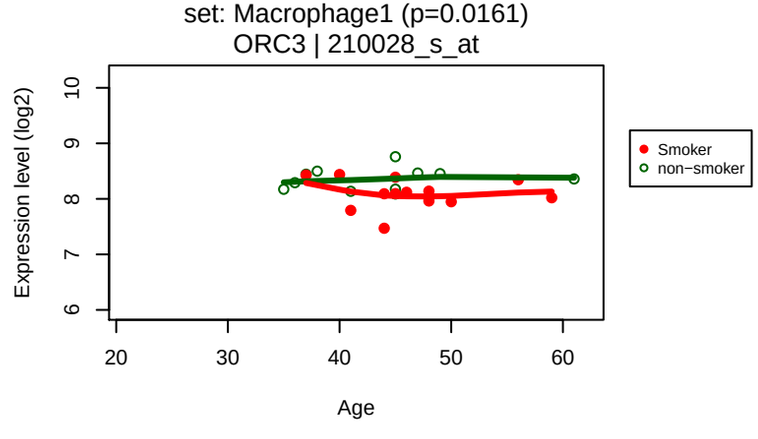

set: SmallAirway1 (p=0.5006)
ORC3 | 210028_s_at
Expression level (log2)
Age
Smoker
non−smoker

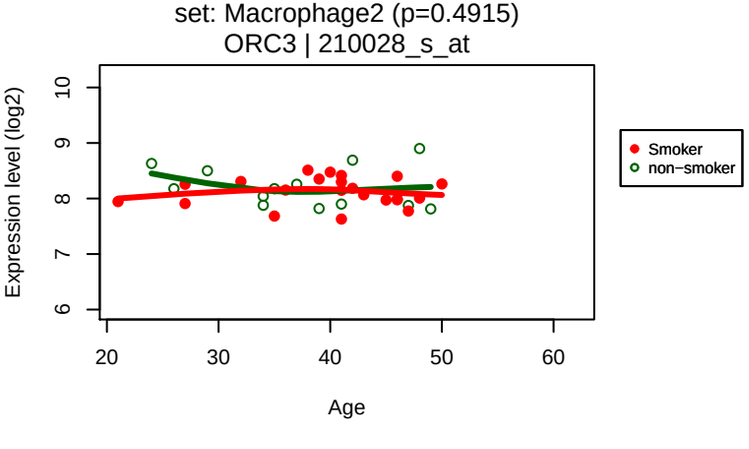

set: SmallAirway2 (p=0.9347)
ORC3 | 210028_s_at
Expression level (log2)
Age
Smoker
non−smoker

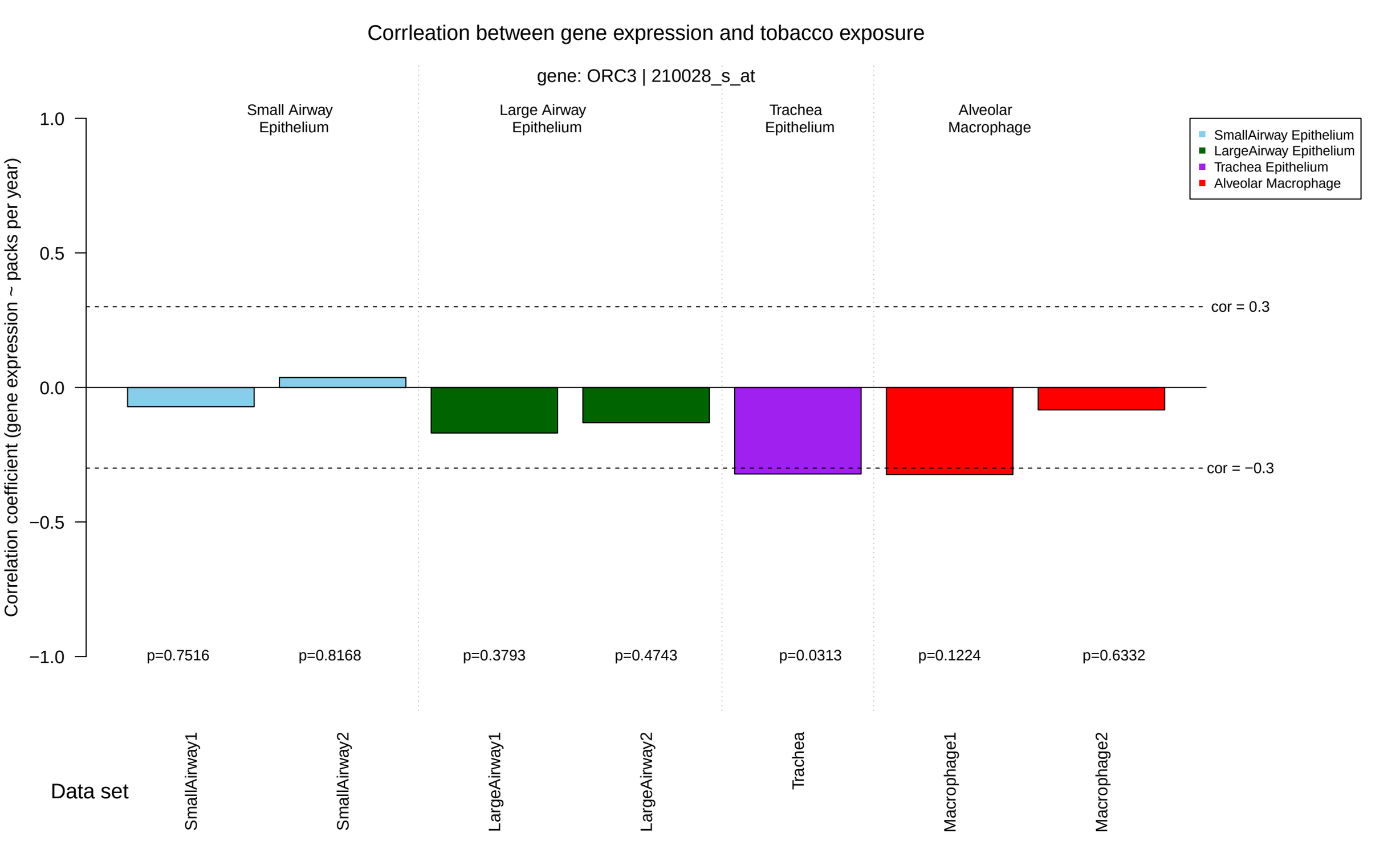

set: LargeAirway1 (p=0.0198)
ORC3 | 210028_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0693)
ORC3 | 210028_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.003)
ORC3 | 210028_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0161)
ORC3 | 210028_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4915)
ORC3 | 210028_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ORC3 | 210028_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7516
p=0.8168
p=0.3793
p=0.4743
p=0.0313
p=0.1224
p=0.6332
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2